Car Price Prediction - ML Report

1. Objective

The goal of this project is to predict used car prices using regression models and analyze feature importance.

2. Data Preprocessing

Performed feature engineering, handled categorical variables using one-hot encoding, treated outliers, and scaled numerical features.

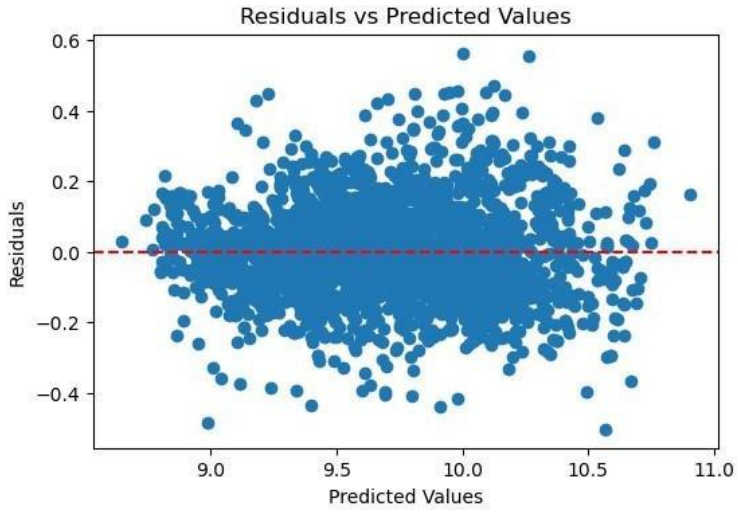
The dataset underwent extensive preprocessing to ensure high-quality inputs for model training. Initially, categorical variables such as fuel type, body type, and transmission were transformed using one-hot encoding. Complex multi-value columns like Comfort_Convenience, Entertainment_Media, Extras, and Safety_Security were engineered into numerical features by counting the number of features present (e.g., num_comfort, num_media).

New features such as km_per_year and power_per_cc were created to capture more meaningful relationships. Highly correlated and redundant features (e.g., Displacement_cc, power_per_cc, and total_features) were removed based on VIF analysis to reduce multicollinearity.

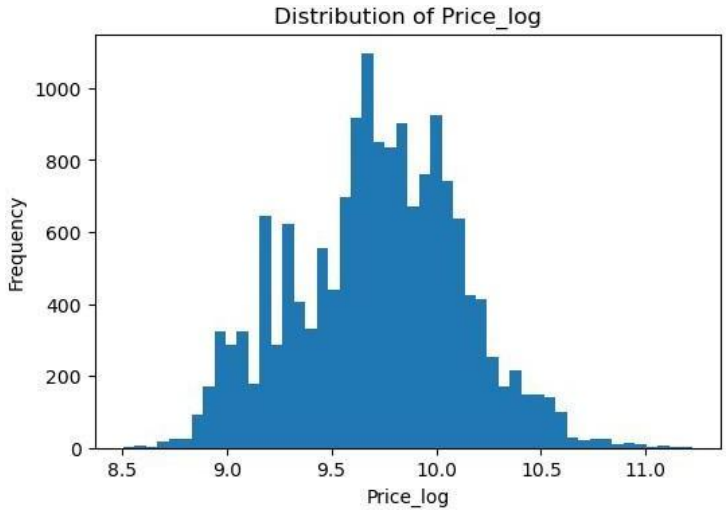
Outliers were identified using statistical methods and treated appropriately to prevent distortion in model learning. The target variable (price) was log-transformed to reduce skewness and improve normality, as seen in the distribution plot. Finally, numerical features were scaled using MinMaxScaler after train-test splitting to avoid data leakage.

3. Models Used

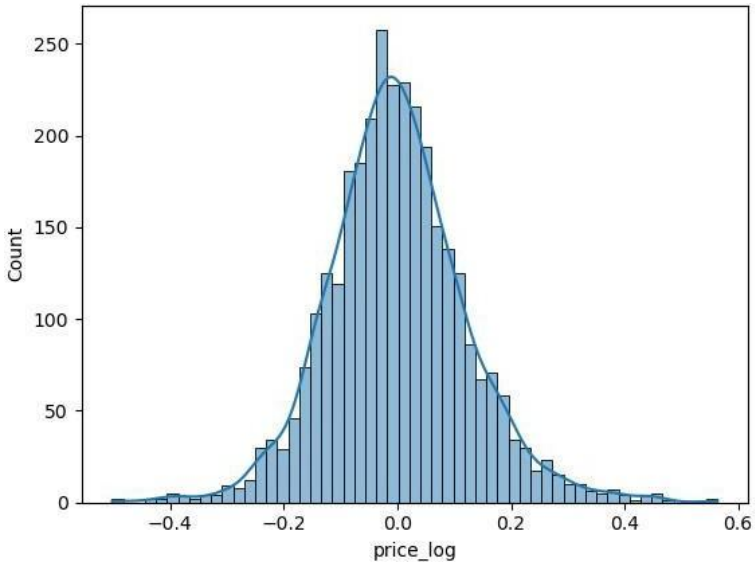
Three models were implemented: Linear Regression, Ridge Regression, and Lasso Regression.



PRICE AFTER LOG TRANSFORMATION



NORMALITY IN RESIDUAL DISTRIBUTION



4. Hyperparameter Tuning

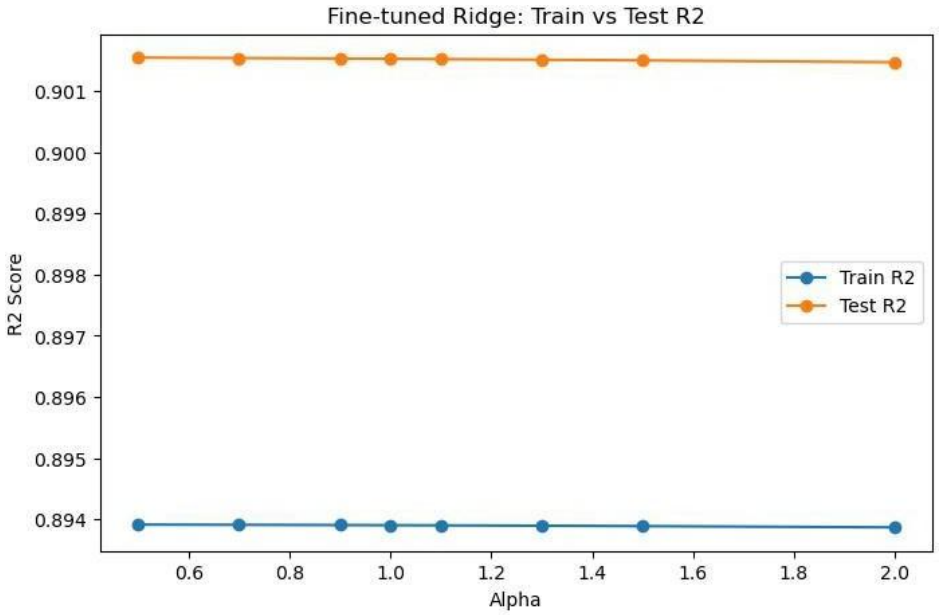
Ridge best alpha: 2. Lasso best alpha: 0.0002.

Hyperparameter tuning was performed to identify the optimal regularization strength for Ridge and Lasso regression models. Initially, a coarse grid of alpha values was tested to identify the approximate optimal range. This was followed by fine-tuning within a narrower range around the best-performing values.

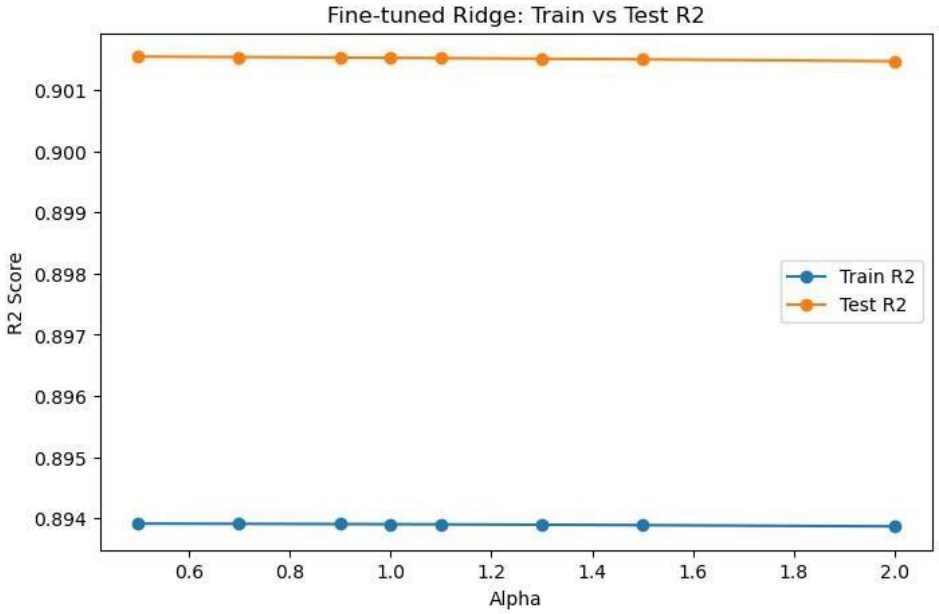
For Ridge regression, the optimal alpha value was found to be 2, indicating that moderate regularization helped stabilize coefficients without significantly reducing model performance. For Lasso regression, the optimal alpha was 0.0002, suggesting that only minimal regularization was required and most features were already relevant.

Cross-validation using negative mean absolute error (neg MAE) was used as the scoring metric to ensure robust model selection. The tuning process confirmed that excessive regularization leads to underfitting, while very small alpha values maintain performance similar to linear regression.

FINE TUNED RIDGE



LASSO FINE TUNED



5. Model Performance

All three models—Linear Regression, Ridge, and Lasso—achieved very similar predictive performance, with R^2 values close to 0.90, indicating that each model was able to explain a high proportion of variance in car prices. This suggests that the

underlying relationships between features and the target variable are largely linear and well captured even by the baseline model. Ridge regression demonstrated slightly better stability by shrinking coefficients and reducing the impact of multicollinearity, which led to more consistent and reliable predictions across different data splits. On the other hand, Lasso regression, with a very small optimal alpha value, performed minimal feature selection and retained most of the predictors, indicating that the majority of features were relevant and contributed meaningfully to the model. Overall, the similarity in performance highlights that the dataset was well-preprocessed, and no model had a significant advantage in terms of predictive accuracy.

6. Key Insights and features

Price tends to increase with higher engine power, greater comfort features, and larger vehicle weight, as these characteristics are often associated with premium or higher-end vehicles. Cars with more powerful engines typically deliver better performance and driving experience, which justifies a higher market price. Similarly, an increased number of comfort features—such as climate control, advanced infotainment systems, and driver assistance technologies—enhances the overall user experience and perceived value of the vehicle. Heavier vehicles often belong to larger segments like SUVs or luxury sedans, which naturally command higher prices due to their build quality, space, and additional features.

On the other hand, price decreases with higher usage and age. Vehicles that have been driven more frequently (higher kilometers per year) experience greater wear and tear, reducing their resale value. Likewise, older cars depreciate over time due to factors such as outdated technology, reduced efficiency, and increased maintenance requirements, leading to lower market prices.

Feature importance analysis revealed that variables such as engine power (hp_kW), usage (km_per_year), comfort features (num_comfort), and vehicle weight (Weight_kg) had the strongest influence on price prediction.

Lasso regression was used for feature selection by shrinking less important feature coefficients to zero. However, due to the small optimal alpha value, only a few features were eliminated, indicating that most features were informative and contributed to the model.

Additionally, multicollinearity analysis using VIF led to the removal of redundant features such as Displacement_cc and total_features, improving model stability. Ridge regression further helped by shrinking correlated feature coefficients, ensuring more reliable and interpretable results.

.

7. Conclusion

The linear regression model was found to be sufficient for this problem, as it already achieved strong performance with an R^2 score of around 0.90, indicating that it explained a large portion of the variance in car prices. Regularisation techniques such as Ridge and Lasso did not lead to significant improvements in performance because the dataset was already well-preprocessed, with proper feature engineering, scaling, and handling of multicollinearity. Overfitting was not observed, as the train and test R^2 scores were very close, showing that the model generalized well to unseen data. Evaluation metrics such as Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and R^2 were used to assess performance, and all three models produced similar results across these metrics. The optimal alpha values were relatively small (Ridge ≈ 2, Lasso ≈ 0.0002), indicating that only mild regularisation was needed. This further confirms that the base linear model was already stable and effective, and additional regularisation provided only marginal benefits.